Matrix identities as derivatives of determinant identities

Terence Tao · 13 January 2013

https://terrytao.wordpress.com/2013/01/13/matrix-identities-as-derivatives-of-determinant-identities/

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The determinant $\det(A)$ of a square matrix A obeys a large number of important identities, the most basic of which is the multiplicativity property

$$\det(AB) = \det(A)\det(B) \tag{1}$$

whenever A, B are square matrices of the same dimension. This identity then generates many other important identities. For instance, if A is an $n \times m$ matrix and B is an $m \times n$ matrix, then by applying the previous identity to equate the determinants of $\begin{pmatrix} 1 & -A \\ B & 1 \end{pmatrix}\begin{pmatrix} 1 & A \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & A \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & -A \\ B & 1 \end{pmatrix}$ (where we will adopt the convention that 1 denotes an identity matrix of whatever dimension is needed to make sense of the expressions being computed, and similarly for 0) we obtain the Weinstein-Aronszajn determinant identity

$$\det(1 + AB) = \det(1 + BA). \tag{2}$$

This identity, which converts an $n \times n$ determinant into an $m \times m$ determinant, is very useful in random matrix theory (a point emphasised in particular by Deift), particularly in regimes in which m is much smaller than n.

Another identity generated from (1) arises when trying to compute the determinant of a $(n+m) \times (n+m)$ block matrix

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

where A is an $n \times n$ matrix, B is an $n \times m$ matrix, C is an $m \times n$ matrix, and D is an $m \times m$ matrix. If A is invertible, then we can manipulate this matrix via block Gaussian elimination as

$$\begin{aligned} \begin{pmatrix} A & B \\ C & D \end{pmatrix} &= \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & A^{-1}B \\ C & D \end{pmatrix} \\ &= \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ C & 1 \end{pmatrix}\begin{pmatrix} 1 & A^{-1}B \\ 0 & D - CA^{-1}B \end{pmatrix} \end{aligned}$$

and on taking determinants using (1) we obtain the *Schur determinant identity*

$$\det\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(A)\det(D - CA^{-1}B) \tag{3}$$

relating the determinant of a block-diagonal matrix with the determinant of the Schur complement $D - CA^{-1}B$ of the upper left block A. This identity can be viewed as the correct way to generalise the 2×2 determinant formula

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc = a(d - ca^{-1}b).$$

It is also possible to use determinant identities to deduce other matrix identities that do not involve the determinant, by the technique of matrix differentiation (or equivalently, matrix linearisation). The key observation is that near the identity, the determinant behaves like the trace, or more precisely one has

$$\det(1+\varepsilon A) = 1 + \varepsilon\ \mathrm{tr}(A) + O(\varepsilon^2) \tag{4}$$

for any bounded square matrix A and infinitesimal ε. (If one is uncomfortable with infinitesimals, one can interpret this sort of identity as an asymptotic as $\varepsilon \to 0$.) Combining this with (1) we see that for square matrices A, B of the same dimension with A invertible and A^{-1}, B invertible, one has

$$\begin{aligned}\det(A+\varepsilon B) &= \det(A)\det(1+\varepsilon A^{-1}B)\\ &= \det(A)(1+\varepsilon\ \mathrm{tr}(A^{-1}B) + O(\varepsilon^2))\end{aligned}$$

for infinitesimal ε. To put it another way, if $A(t)$ is a square matrix that depends in a differentiable fashion on a real parameter t, then we have the Jacobi formula

$$\frac{d}{dt}\det(A(t)) = \det(A(t))\ \mathrm{tr}(A(t)^{-1}\frac{d}{dt}A(t))$$

whenever $A(t)$ is invertible. (Note that if one combines this identity with cofactor expansion, one recovers Cramer's rule.)

Let us see some examples of this differentiation method. If we take the Weinstein-Aronszajn identity (2) and multiply one of the rectangular matrices A by an infinitesimal ε, we obtain

$$\det(1+\varepsilon AB) = \det(1+\varepsilon BA);$$

applying (4) and extracting the linear term in ε (or equivalently, differentiating at ε and then setting $\varepsilon = 0$) we conclude the cyclic property of trace:

$$\mathrm{tr}(AB) = \mathrm{tr}(BA).$$

To manipulate derivatives and inverses, we begin with the Neumann series approximation

$$(1+\varepsilon A)^{-1} = 1 - \varepsilon A + O(\varepsilon^2)$$

for bounded square A and infinitesimal ε, which then leads to the more general approximation

$$\begin{aligned}(A+\varepsilon B)^{-1} &= (1+\varepsilon A^{-1}B)^{-1}A^{-1}\\ &= A^{-1} - \varepsilon A^{-1}BA^{-1} + O(\varepsilon^2)\end{aligned} \tag{5}$$

for square matrices A, B of the same dimension with B, A^{-1} bounded. To put it another way, we have

$$\frac{d}{dt}A(t)^{-1} = -A(t)^{-1}(\frac{d}{dt}A(t))A(t)^{-1}$$

whenever $A(t)$ depends in a differentiable manner on t and $A(t)$ is invertible.

We can then differentiate (or linearise) the Schur identity (3) in a number of ways. For instance, if we replace the lower block D by $D+\varepsilon H$ for some test $m \times m$ matrix H, then by (4), the left-hand side of (3) becomes (assuming the invertibility of the block matrix)

$$(\det\begin{pmatrix}A & B\\ C & D\end{pmatrix})(1+\varepsilon\ \mathrm{tr}\begin{pmatrix}A & B\\ C & D\end{pmatrix}^{-1}\begin{pmatrix}0 & 0\\ 0 & H\end{pmatrix} + O(\varepsilon^2))$$

while the right-hand side becomes

$$\det(A)(\det(D-BA^{-1}C) + \varepsilon\ \mathrm{tr}((D-BA^{-1}C)^{-1}H) + O(\varepsilon^2));$$

extracting the linear term in ε, we conclude that

$$\mathrm{tr}\ (\begin{pmatrix}A & B\\ C & D\end{pmatrix}^{-1}\begin{pmatrix}0 & 0\\ 0 & H\end{pmatrix}) = \mathrm{tr}((D-BA^{-1}C)^{-1}H).$$

As H was an arbitrary $m \times m$ matrix, we conclude from duality that the lower right $m \times m$ block of $\begin{pmatrix}A & B\\ C & D\end{pmatrix}^{-1}$ is given by the inverse $(D-BA^{-1}C)^{-1}$ of the Schur complement:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} ?? & ?? \\ ?? & (D - BA^{-1}C)^{-1} \end{pmatrix}.$$

One can also compute the other components of this inverse in terms of the Schur complement $D - BA^{-1}C$ by a similar method (although the formulae become more complicated). As a variant of this method, we can perturb the block matrix in (3) by an infinitesimal multiple of the identity matrix giving

$$\det \begin{pmatrix} A+\varepsilon & B \\ C & D+\varepsilon \end{pmatrix} = \det(A+\varepsilon)\det(D+\varepsilon - C(A+\varepsilon)^{-1}B). \tag{6}$$

By (4), the left-hand side is

$$(\det \begin{pmatrix} A & B \\ C & D \end{pmatrix})(1 + \varepsilon \operatorname{tr} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} + O(\varepsilon^2)).$$

From (5), we have

$$D + \varepsilon - C(A+\varepsilon)^{-1}B = D - CA^{-1}B + \varepsilon(1 + CA^{-2}B) + O(\varepsilon^2)$$

and so from (4) the right-hand side of (6) is

$$\det(A)\det(D - CA^{-1}B) \times$$

$$\times (1 + \varepsilon(\operatorname{tr}(A^{-1}) + \operatorname{tr}((D - CA^{-1}B)^{-1}(1 + CA^{-2}B))) + O(\varepsilon^2));$$

extracting the linear component in ε, we conclude the identity

$$\operatorname{tr} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \operatorname{tr}(A^{-1}) + \operatorname{tr}((D - CA^{-1}B)^{-1}(1 + CA^{-2}B)) \tag{7}$$

which relates the trace of the inverse of a block matrix, with the trace of the inverse of one of its blocks. This particular identity turns out to be useful in random matrix theory; I hope to elaborate on this in a later post.

As a final example of this method, we can analyse low rank perturbations $A + BC$ of a large $(n \times n)$ matrix A, where B is an $n \times m$ matrix and C is an $m \times n$ matrix for some $m < n$. (This type of situation is also common in random matrix theory, for instance it arose in this previous paper of mine on outliers to the circular law.) If A is invertible, then from (1) and (2) one has the matrix determinant lemma

$$\det(A + BC) = \det(A)\det(1 + A^{-1}BC) = \det(A)\det(1 + CA^{-1}B);$$

if one then perturbs A by an infinitesimal matrix εH, we have

$$\det(A + BC + \varepsilon H) = \det(A + \varepsilon H)\det(1 + C(A + \varepsilon H)^{-1}B).$$

Extracting the linear component in ε as before, one soon arrives at

$$\operatorname{tr}((A + BC)^{-1}H) = \operatorname{tr}(A^{-1}H) - \operatorname{tr}((1 + CA^{-1}B)^{-1}CA^{-1}HA^{-1}B)$$

assuming that A and $A + BC$ are both invertible; as H is arbitrary, we conclude (after using the cyclic property of trace) the Sherman-Morrison formula

$$(A + BC)^{-1} = A^{-1} - A^{-1}B(1 + CA^{-1}B)^{-1}CA^{-1}$$

for the inverse of a low rank perturbation $A + BC$ of a matrix A. While this identity can be easily verified by direct algebraic computation, it is somewhat difficult to *discover* this identity by such algebraic manipulation; thus we see that the "determinant first" approach to matrix identities can make it easier to find appropriate matrix identities (particularly those involving traces and/or inverses), even if the identities one is ultimately interested in do not involve determinants. (As differentiation typically makes an identity lengthier, but also more "linear" or "additive", the determinant identity tends to be shorter (albeit more nonlinear and more multiplicative) than the differentiated identity, and can thus be slightly easier to derive.)

> **Exercise 1** Use the "determinant first" approach to derive the Woodbury matrix identity (also known as the binomial inverse theorem)
>
> $$(A+BDC)^{-1} = A^{-1} - A^{-1}B(D^{-1}+CA^{-1}B)^{-1}CA^{-1}$$
>
> where A is an $n \times n$ matrix, B is an $n \times m$ matrix, C is an $m \times n$ matrix, and D is an $m \times m$ matrix, assuming that A, D and $A+BDC$ are all invertible.

> **Exercise 2** Let A, B be invertible $n \times n$ matrices. Establish the identity
>
> $$\det(A+B)\det(A-B) = \det(B)\det(AB^{-1}A - B)$$
>
> and differentiate this in A to deduce the identity
>
> $$(A+B)^{-1} + (A-B)^{-1} = 2(A - BA^{-1}B)^{-1}$$
>
> (assuming that all inverses exist) and thence
>
> $$(A+B)^{-1} = (A - BA^{-1}B)^{-1} + (B - AB^{-1}A)^{-1}.$$
>
> Rotating B by i then gives
>
> $$(A+iB)^{-1} = (A + BA^{-1}B)^{-1} - i(B + AB^{-1}A)^{-1},$$
>
> which is useful for inverting a matrix $A+iB$ that has been split into a self-adjoint component A and a skew-adjoint component iB.

Comments (45)



Mark Meckes · 13 January 2013

Thanks! I'm about to start teaching a matrix analysis course in which I was going to cover most of this material, though I hadn't thought of quite this viewpoint on the second half. I think you've just saved me the work of preparing one lecture.



Mark Meckes · 31 January 2013

I'm also delighted to discover that telling my browser to print this page results in a printout of the content of the post without all the links on the left.

José Carlos Santos · 13 January 2013

When you wrote "{n+m \times n+m} square matrices", what you meant was "{(n+m) \times (n+m)} square matrices".
[Parentheses added – T.]



Matrix identities as derivatives of determinant identities « Guzman's Mathematics Weblog · 13 January 2013

[...] via Matrix identities as derivatives of determinant identities. [...]



Anonymous · 13 January 2013

Should the matrix [1,C;0,1] above equation (3) be [1,0;C,1]?
[Corrected, thanks – T.]



Joseph Nebus · 13 January 2013

I think I'm most startled to realize that I don't remember having encountered Sylvester's determinant identity before. It seems like a wonderful little bit to show off. (But I suppose everyone's life is filled with wonderful little bits they run across only surprisingly late.)

******DK** · 13 January 2013

Hello, I'm a big fan of Terry Tao's "What's new" blog, however mainland China has blocked the WordPress. Therefore, I would appreciate it very much if you could send me full copy of each new post better in PDF rather than just e-mail since the character won't show up.
Send from Ted

I am not an anoymous · 6 February 2013

I use freegate + Autoproxy(a plugin of firefox) to circumvent the GFW,you can download freegate here:http://forums.internetfreedom.org/index.php?topic=18444.0
Also,there are several other ways to circumvent the great firewall of China,there are some discussion here:

> WordPress blocked again by great firewall of China?

Anonymous · 14 January 2013

Hi,
Is there a minus missing on the right-hand side of the equation below (5)? (the equation of the derivative of A(t) inverse)
[Corrected, thanks – T.]

Peter · 14 January 2013

One identity that has been useful for me recently involves two matrices $A, B \in SL_2(\mathbb{C})$. The identity is
$tr(A)tr(B) = tr(AB) + tr(AB^{-1})$
Out of curiosity, do you know if this can be derived using the methods you described? The identity isn't hard to prove using the fact that a matrix satisfies its own characteristic polynomial, but I don't know a very satisfying proof.

Terence Tao · 14 January 2013

Hmm, that is a puzzle. It can be deduced by differentiating the 2×2 matrix identity
$\det(A+B) = \det(A) + \det(B) + \mathrm{tr}(A)\ \mathrm{tr}(B) - \mathrm{tr}(AB)$, (*)
which is valid for arbitrary 2×2 matrices A, B, in the A direction (at $A = 0$), but then I don't know of a good reason (other than brute force calculation) as to why (*) holds, other than an observation that five quantities $\det(A), \det(B), \mathrm{tr}(A), \mathrm{tr}(B), \mathrm{tr}(AB)$ parameterise the $8 - 3 = 5$-dimensional space of pairs (A, B) of 2×2 matrices up to conjugation, so *some* identity of the form (*) ought to exist, and then degree considerations indicate that it should have the homogeneous structure indicated in (*).

Anonymous · 1 February 2014

The identity det(X + Y) = det(X) + det(Y) – tr(XY) + tr(X)tr(Y) is a special case of the identity described in Reutenauer and Schuetzenberger's paper giving the determinant of the sum of k square matrices of order n as a sum of k^n terms. For example, if X and Y are 3×3 matrices then det(X + Y) = det(X) + det(Y) – tr(XY)tr(X) – tr(XY)tr(Y) + c(X)tr(Y) + tr(X)c(Y) + tr(XXY) + tr(XYY) , where c(X) = (tr(X)^2 – tr(X^2)) / 2.

Gil Kalai · 18 January 2013

One thing that comes to mind in this context are polynomial identities for the ring of matrices like the Amitsur-Levitzki identity (See, e.g. http://gilkalai.wordpress.com/2009/05/12/the-amitsur-levitski-theorem-for-a-non-mathematician/) For those there is a result by Processi and by Razmislov which

asserts that all PI identities are derived from the Cayley-Hamilton theorem, which is a sort of determinant-identity.

> **A researcher** · 18 January 2013
>
> Hello Gil, would you be able to give us a suitable source for the Processi or Razmislov works? Further, could you explain what do you mean by "all PI identities are derived from the C-H theorem"? I thought that to find the generating set of all PI identities of the ring of matrices is an open problem. Only for 2×2 matrices it was solved by Drenski in 1981. But maybe we're talking about different definitions?
>
> > **Gil Kalai** · 19 January 2013
> >
> > Perhaps for the results I mentioned "are derived" is quite a description of generating sets. There is an article "Polynomial identities and the Cayley-Hamilton theorem" by Edward Formanek in The Mathematical Intelligencer 11(89) 37-39 which describes the related works of Processi and Razmislov. Rowen's BAMS book review of a book by Formanek and Drensky http://www.ams.org/bull/2006-43-02/S0273-0979-06-01082-2/S0273-0979-06-01082-2.pdf contains a lot of interesting information.

Nick Higham · 19 January 2013

There are lots of nice result along these lines. One of my favorites is $Af(BA) = f(AB)A$ for any $m \times n$ A and $n \times m$ B such that both matrix functions (where f maps square matrices to square matrices) are defined. In particular, $A(BA)^{1/2} = (AB)^{1/2}A$. The proofs are easy using the theory of matrix functions (see Section 1.8 of http://www.maths.manchester.ac.uk/~higham/fm/index.php).

Anonymous · 21 January 2013

As usual, a wonderful to learn from exposition. I was thinking if Prof. Tao could have also related determinant (det) to the rank function. I mean, the current literature suggests det as a good surrogate for the rank, than, for example, the trace function. Probably, this could have motivated some tighter approximations for the rank operator which can be immensely useful, for instance, in numerical optimization. Of course, other application for this very aspect arises in compressive sensing, and, Prof. Tao is again a seminal contributor in this area of signal processing!

Amit Bhaya · 30 January 2013

There's also the lovely and well known formula \det \exp A = \exp trace(A) with its connection to Liouville's theorem on conservation of phase space volume by Hamiltonian flows (since Hamiltonian matrices have zero trace).

teobanica · 31 January 2013

Hello, I'm teaching some linear algebra this semester, and, as always, quite hard to get through the first definitions, as to be able to start speaking about the magics of det.. There's actually this problem with the very definition of $\det(A)$: is that a volume (intuitive and nice and everything), or some more abstract thing? I learned in Romania, using abstract stuff, now I'm teaching in France, and I have to use abstract stuff as well (otherwise I'd be probably put on the death row by my colleagues :) On the other hand a Russian colleague of mine told me that, at least some 20 years ago, $\det(A)$ was by definition a volume, in Moscow! (that's probably why Russians are so good at it) I was wondering how the situation in the US is: any choice allowed? For math students of course.

anonymous · 5 February 2013

Dear Professor Tao,

I did the calculation of the products of the two matrices X and Y, and it does not seem to give the Sylvester determinant equaltiy. Could this point be further elaborated on?



Terence Tao · 5 February 2013

Oops, I didn't set up the matrices correctly. It should be OK now…



Mike · 23 July 2013

If I'm not missing something, I think you really wanted the second product to simply be the first one in reverse order, instead of using the matrix with a 0 and a -B.
[Corrected, thanks – T.]

Artificial Intelligence Blog · "Matrix identities as derivatives of determinant identities" · 14 February 2013

[…] Check out Terence Tao's wonderful post "Matrix identities as derivatives of determinant identities". […]



Craig Tracy · 17 February 2013

For an account of the use of the Sylvester det identity, det(I+AB)=det(I+BA), in random matrix theory, see C. Tracy & H. Widom, "Correlation functions, cluster functions and spacing distributions for random matrices", J. of Statistical Physics 92 (1998) 809-835.
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Supercommutative gaussian integration, and the gaussian unitary ensemble « What's new · 19 February 2013

[…] later in this post. By the trick of matrix differentiation of the determinant (as reviewed in this recent blog post), one can also use this method to compute matrix-valued statistics such […]

Study Hacks » Blog Archive » You Can Be Busy or Remarkable — But Not Both · 3 April 2013

[…] new year, he's written nine long posts, full of mathematical equations and fun titles, like "Matrix identities as derivatives of determinant identities." His most recent post is 3700 words long! And that's a normal […]



Kyle Kloster · 15 January 2014

I'm having trouble with the derivation of the block inverse form, above (6):
the term $CA^{-1}B$ in (3) now appears as $BA^{-1}C$, and I can't figure out why. Then, equating the linear terms in ϵ is said to yield
$\mathrm{tr}([A,B;C,D]^{-1}[0,0;0,H]) = \mathrm{tr}((D-BA^{-1}C)^{-1}H)$
I don't understand what happened to the other coefficients of these linear terms, $\det(A)$ for the right-hand side, and $\det([A,B;C,D])$ for the left-hand side.
What am I missing?



Terence Tao · 15 January 2014

Oops, there were some typos in the text (in particular, all occurrences of $BA^{-1}C$, which doesn't even make sense as a matrix product, should be $CA^{-1}B$ instead); hopefully they are corrected now.



Kyle Kloster · 15 January 2014

Thanks for the response!
There is still one occurrence of $BA^{-1}C$, in the paragraph immediately prior to equation (6).
[Corrected, thanks – T.]

Anonymous · 15 January 2014

Typo: Exercise 2: “thence” –> “hence”.
[Corrected, thanks – T.]

My Math Diary · 20 April 2014

[...] Derivative of determinant: Terry Tao, Stack [...]

Derived multiplicative functions | What's new · 24 September 2014

[...] identities, such as (1). This phenomenon is analogous to the one in linear algebra discussed in this previous blog post, in which many of the trace identities used there are derivatives of determinant identities. For [...]

Pascal · 3 April 2017

Hi, may i know is there any analytic formula for the derivative of adjugate matrix w.r.t itself or its elements? i.e.
$\frac{\partial \operatorname{adj}(A)}{\partial A}$
or
$\frac{\partial \operatorname{adj}(A)}{\partial a_{ij}}$

Quantitative continuity estimates | What's new · 22 May 2017

[...] is a scalar such that is non-zero. (See also this previous blog post for more discussion of these sorts of [...]

起源笔记 / 基于云数据的网络松鼠症分布模型设计一文 · 4 June 2017

[...] a scalar such that is non-zero. (See also this previous blog post for more discussion of these sorts of [...]

user777 · 23 July 2017

Is it possible to have any such derivations for permanent-trace or permanent-determinant?

Dodgson condensation from Schur complementation | What's new · 28 August 2017

[...] can in turn be used to establish many further identities; in particular, as shown in A HREF=”https://terrytao.wordpress.com/2013/01/13/matrix-identities-as-derivatives-of-determinant-identities… previous post, it implies the Schur determinant [...]

Inverting the Schur complement, and large-dimensional Gelfand-Tsetlin patterns | What's new · 16 September 2017

[...] the other hand, by the Woodbury matrix identity (discussed in this previous blog post), we [...]

Anonymous · 17 October 2017

Is there any book which discusses this stuff and particularly identity 4 and its corresponding identity

Chris · 1 August 2018

how can I find the derivative of det(I-2itAΣ) ? Please help!

DREAM PETER SARNAK 21 AUGUST 2018 | zulfahmed · 21 August 2018

[...] what is necessary to understand is the relation between determinant and trace and Terry has a very nice article clarifying these issues so we can understand the relations clearly: “The key observation is [...]

Leonardo Torres · 13 April 2019

What is a reference for (4)?

 **Anonymous** · 13 April 2019

characteristic polynomial

 **Gilad Lerman** · 12 November 2021

Typo: In Exercise 2 the minus sign between the two inverses in the last two equations should be a plus sign (just found these notes when searching for material relevant for some teaching notes I write and I like it)

[Minus sign in the third equation corrected; the minus sign in the fourth equation however appears to be correct – T.]

 **Gilad Lerman** · 12 November 2021

yes, of course (I focused on the third line)